PackComply AI

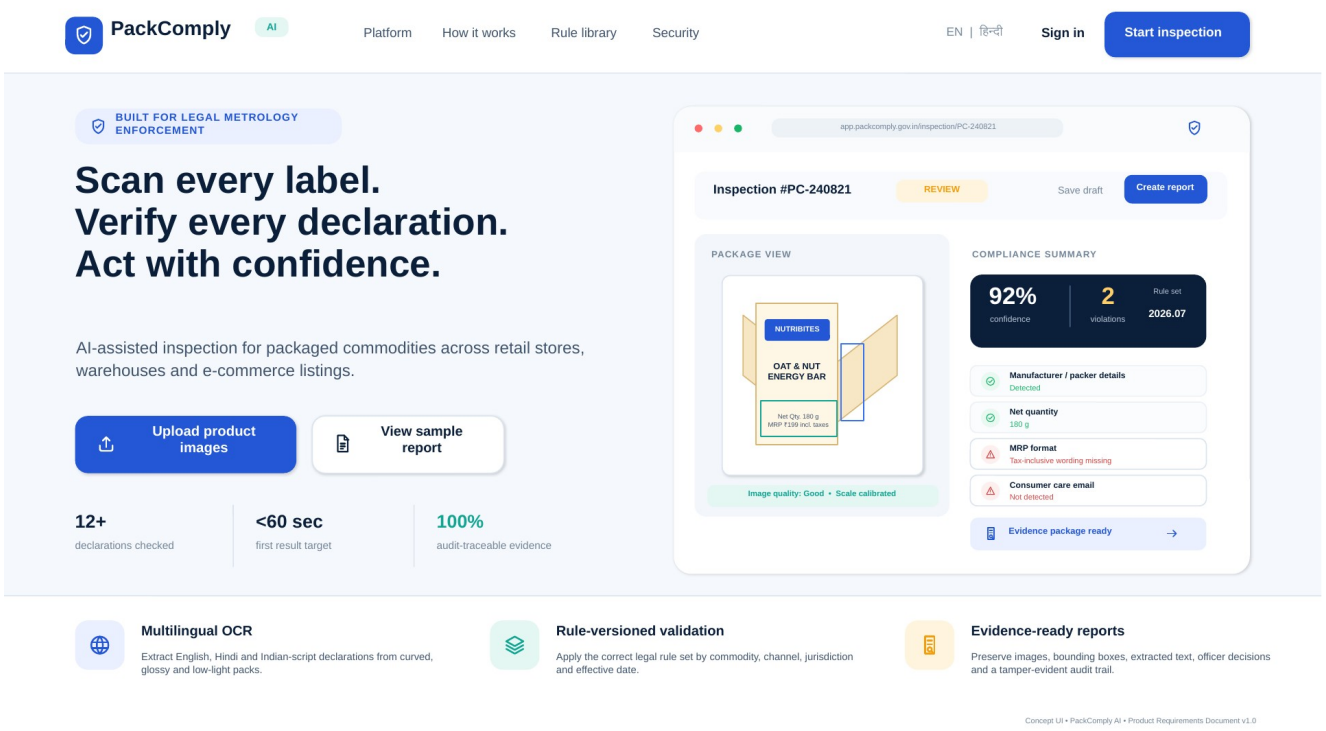
PRODUCT REQUIREMENTS DOCUMENT

PackComply AI

Automated Legal Metrology Compliance Inspection Platform

Scan packaged commodity labels and e-commerce listings, extract statutory declarations, evaluate the applicable rule version and produce audit-ready evidence for enforcement officials.

AI-assisted | **Rule-versioned** | **Evidence-first**



Version	1.0	**Status**	Proposed / for stakeholder review
Date	29 August 2026	**Product owner**	To be nominated

FRONT MATTER

Document control and use

Field	Definition
Document purpose	Define the product, legal-rule architecture, workflows, functional and non-functional requirements, delivery scope, acceptance criteria and implementation roadmap.
Primary audience	Product owners, Legal Metrology officials, legal reviewers, UX and engineering teams, data science teams, security teams, implementation partners and UAT participants.
Baseline date	29 August 2026. Statutory and departmental changes after this date must be added as a new rule version before use.
Decision authority	The software provides decision support. The authorized enforcement officer remains responsible for the final finding, notice, prosecution or closure decision.
Change control	Every PRD change, legal rule change, model change and report-template change requires versioning, approval and a retained audit record.

Legal and operational disclaimer

The Legal Metrology Act, the applicable Packaged Commodities Rules, Gazette notifications, sector-specific laws, official directions and competent legal review remain authoritative. Automated results must not be represented as a final adjudication, especially when image quality, physical scale, product classification or legal applicability is uncertain.

Approval matrix

Role	Approval responsibility	Status
Product owner	Scope, priorities, success metrics	Pending
Legal Metrology domain owner	Rule interpretation and legal test pack	Pending
Security authority	Security architecture and production clearance	Pending
Engineering lead	Technical feasibility and estimates	Pending
UAT authority	Acceptance evidence and go-live recommendation	Pending

Contents

Section	Topic
01	Executive summary
02	Background, problem and product vision
03	Objectives, non-goals and success measures
04	Users, roles and permissions
05	Scope and phased delivery
06	User journeys and operating workflows
07	Regulatory requirements and rule model

Section	Topic
08	Functional requirements
09	AI, OCR and computer vision requirements
10	User experience, accessibility and localization
11	Data model and retention
12	Solution architecture and deployment
13	Security, privacy and evidence integrity
14	Non-functional requirements
15	Reports, dashboards and analytics
16	Testing, acceptance and release gates
17	Roadmap, risks and governance
18	Traceability matrix
Appendix A	Landing-page concept
Appendix B	Glossary and official references

01

Executive summary

An enforcement-centric platform for fast, consistent and explainable packaged-commodity inspections.

Challenge Manual inspection is slow and inconsistent across high product volumes, varied packaging, multiple scripts, curved labels and rapidly changing legal requirements.	**Product** A responsive web and mobile platform that captures product evidence, extracts declarations, selects the applicable legal rule version and presents reviewable findings.
Control model AI proposes; deterministic rules evaluate; authorized officers review, correct, override and sign. Every conclusion links to evidence and a legal source.	**Outcome** Reduced inspection effort, better legal consistency, searchable compliance history, stronger evidence packages and risk-based enforcement dashboards.

Product proposition

PackComply AI will support physical-pack inspections, back-office image review and e-commerce listing checks. It will identify declarations, quantify image and extraction confidence, test deterministic compliance rules, preserve the original evidence, and generate a legally reviewable draft report. The platform is designed for central and state enforcement teams, with rule administration controlled by authorized domain officers.

MVP outcome statement

MVP definition of done

An inspector can create an inspection, capture or upload all package views, receive quality guidance, review extracted declarations and rule findings, record an officer decision, attach evidence, and export a signed PDF plus an editable report. Supervisors can search the repository, monitor workload and publish approved rule updates without redeploying application code.

02

Background, problem and product vision

Background

Packaged commodities are sold through retail stores, supermarkets, warehouses and online marketplaces across India. The Legal Metrology Act, 2009 and the Legal Metrology (Packaged Commodities) Rules, 2011 require prescribed declarations and provide enforcement consequences for non-compliant pre-packaged commodities. The declaration set and its presentation depend on product type, package characteristics, sales channel, exemptions, sector-specific laws and the rule version effective on the relevant date. [R1] [R2]

Problem statement

- Inspection volume is substantially larger than the number of products that can be manually checked in depth.
- Declarations may appear across multiple package faces, in multiple scripts, with glare, curvature, low contrast, small type or overprinted stickers.
- Determining applicability requires product classification, package attributes, channel, import status, package date and effective legal amendments.
- Manual findings can lack standardized evidence, consistent rule citations, searchable history or a reproducible audit trail.

- Rule changes can make a static checklist obsolete and can cause historical inspections to be evaluated against the wrong version.

Product vision

> **Vision**
> Enable every authorized Legal Metrology officer to move from product image to an explainable, evidence-linked compliance assessment in minutes - without replacing professional judgment or legal due process.

Design principles

Principle	Application
Evidence before conclusion	Original images, source pages, timestamps and hashes are retained before processing.
Legal rules as versioned data	Effective dates, applicability and citations are configurable, approved and testable.
Human authority	The officer can inspect source evidence, correct extraction and record a reasoned override.
Uncertainty is explicit	The system uses Not Assessable and Needs Review rather than forcing a false pass/fail.
Open and interoperable	Reports, evidence and structured results are exportable without vendor lock-in.
Security by design	Least privilege, jurisdictional scoping, tamper evidence and full audit logging are default.

03

Objectives, non-goals and success measures

Product objectives

Objective	Outcome
O1	Reduce time spent locating and transcribing declarations.
O2	Improve consistency of legal-rule selection and finding explanations.
O3	Detect missing, incomplete, contradictory, unreadable or improperly presented declarations.
O4	Create standardized digital evidence and editable/PDF compliance reports.
O5	Maintain searchable product, inspection, violation and corrective-action history.
O6	Give managers risk, workload and repeat-non-compliance visibility.
O7	Support legal amendments, category overlays and historical replay without software code changes.
O8	Provide secure role-based operation in low-connectivity field conditions.

Non-goals for the initial release

- Replacing the authorized officer, legal notice process or court-admissibility determination.
- Measuring the actual contents or permissible error of a package without approved measurement instruments and procedures.
- Counterfeit detection, product safety testing, laboratory testing or quality certification.
- Full automation of every sectoral regime such as food, medical devices, cosmetics, seeds or pharmaceuticals in the MVP; these are routed as overlays and expanded in phases.

- Unrestricted web scraping where access controls, marketplace terms or technical barriers prohibit automated capture; a manual evidence-upload path remains available.

Success metrics and launch targets

Measure	Definition	Target
Inspection efficiency	Median officer handling time from accepted images to reviewed draft report	At least 40% lower than the agreed manual baseline after stabilization
Declaration detection	Recall for core declarations on the approved English/Hindi benchmark	>=95% on legible images; results segmented by script and package type
Field extraction	Exact or normalized match for quantity, MRP, dates and contact details	>=90% on approved benchmark
Font assessment	Pass/fail accuracy on calibrated, measurable samples	>=90%; uncalibrated samples must be Not Assessable
Traceability	Final findings with linked rule, evidence region, actor and timestamp	100%
Review safety	High-severity automated findings reviewed by an authorized officer	100% before finalization
Availability	Monthly production availability excluding approved maintenance	>=99.5% for MVP
Adoption	Inspections created and finalized digitally in pilot jurisdictions	Target agreed during pilot planning

04

Users, roles and permissions

Primary personas

Persona	Needs
Field inspector	Capture physical packs, receive camera guidance, review findings, add evidence and prepare reports, often with intermittent connectivity.
Review / senior officer	Validate high-risk findings, approve overrides, allocate work, issue or recommend actions and monitor pending cases.
Legal rule administrator	Translate approved law and notifications into versioned rules, attach sources, run regression tests and publish after dual approval.
Compliance analyst	Search historical products, detect patterns, prepare jurisdiction/category analysis and monitor repeat entities.
System administrator	Manage tenants, users, roles, integrations, configuration, security policies and operational health; cannot alter legal outcomes without authorization.
Auditor / oversight user	Read-only access to evidence, rule versions, reports and audit logs within an approved scope.

Role capability matrix

Capability	Inspector	Reviewer	Rule admin	Analyst	Sys admin	Auditor
Create/capture inspection	Yes	Yes	No	No	Config	Read
Edit extracted declarations	Yes	Yes	Test only	No	No	Read

Capability	Inspector	Reviewer	Rule admin	Analyst	Sys admin	Auditor
Override finding	With reason	Yes	No	No	No	Read
Finalize / approve report	Policy based	Yes	No	No	No	Read
Create / publish legal rules	No	No	Draft / dual approval	No	No	Read
Dashboard and analytics	Own scope	Assigned scope	Rule QA	Broad scope	System health	Read
User and integration admin	No	No	No	No	Yes	No
Audit-log access	Own cases	Scope	Rule changes	Scope	Security scope	Approved scope

Segregation of duties

A rule author cannot independently publish the same rule version. A system administrator cannot modify a finalized inspection finding. A report approver must be independent where agency policy requires second-level approval.

05

Scope and phased delivery

Capability scope

Capability	Included functions	Delivery
Inspection intake	Physical package capture, image/PDF upload, batch upload, e-commerce URL/screenshot/feed intake	MVP
Image intelligence	Quality checks, dewarping, package-face and principal-display-panel assistance, text regions	MVP
Declaration extraction	English and Hindi core fields; configurable additional Indian scripts	MVP + Phase 2
Legal validation	Core LMPC declarations, Rule 7 presentation logic, exemptions, versions and citations	MVP
E-commerce validation	Listing declarations, timestamped evidence, physical-vs-online comparison, country-of-origin filter test	MVP + Phase 2
Case workflow	Officer review, evidence, overrides, notices, re-inspection and closure	MVP + Phase 2
Reports and repository	PDF/editable reports, search, product history, exports and audit trail	MVP
Analytics	Operational dashboards, violation trends, repeat patterns and model-quality dashboards	MVP + Phase 2
Advanced surveillance	Marketplace monitoring, risk scoring, duplicate listings and proactive sampling	Phase 3

MVP supported evidence inputs

- JPEG, PNG, HEIC and camera-captured images; multi-page PDF for label artwork or listing evidence.
- Front, back, side, top, bottom and close-up images, with explicit package-face labels.
- Product identifiers: barcode/GTIN where available, SKU, brand, common name, batch/lot, category and responsible entity.
- E-commerce URL, timestamped full-page screenshot and structured listing fields; direct marketplace feed integrations are optional per partner.
- Optional calibration marker or known package dimension for physical font-size estimation.

Category strategy

The MVP implements a general LMPC rule layer plus a configurable category-routing layer. Products governed by additional sectoral rules are not treated as ordinary packages by default; the system routes them to an approved overlay or marks them for specialist review. The architecture must support food, edible oils, pan masala, medical devices, cosmetics, seeds and other regulated categories without changing the core inspection workflow.

06

User journeys and operating workflows



Figure 1. Evidence-first workflow from intake to report, action and searchable history.

Journey A - physical retail or warehouse inspection

1. Inspector creates an inspection, selects location and seller, and records package context.
2. The mobile camera prompts for all relevant faces, close-ups and a scale reference where font-size measurement is required.
3. The quality gate identifies blur, glare, occlusion, incomplete coverage and uncalibrated views before processing.
4. OCR and layout services extract text, declaration candidates, bounding boxes, script, values and confidence.
5. The rule engine selects the effective rules and produces Passed, Failed, Needs Review, Not Applicable or Not Assessable evaluations.
6. The inspector reviews evidence, corrects extraction, records observations, annotates images and resolves each material finding.
7. The system generates a draft report; an authorized officer finalizes or routes it for approval and action.

Journey B - e-commerce listing inspection

8. Officer submits a URL or captures a listing page with timestamp, seller, marketplace and product identifiers.
9. The system preserves the source, screenshot and available structured fields, then extracts declarations shown to the consumer.
10. Channel-specific rules test required listing information, imported-product country-of-origin visibility and, where applicable, the country-of-origin search/filter experience.
11. The officer can compare the listing against physical-pack images and raise discrepancies or missing-information findings.

Journey C - improvement notice and re-inspection

12. Where the applicable legal workflow permits an improvement notice for a first procedural or regulatory non-compliance, the officer selects the approved notice template and cure period. [R8]
13. The case stores service evidence, response documents and the due date; reminders and escalations follow policy.
14. A re-inspection links to the original product and finding, records corrected evidence and closes, escalates or continues the case.

Journey D - legal rule update

15. Rule administrator registers the official source, publication date, effective date, affected categories and superseded provisions.
16. A draft rule version is configured and tested against positive, negative, boundary and historical samples.
17. A second authorized legal approver reviews the machine-readable logic, human explanation and expected outcomes.
18. The version is scheduled or published; existing finalized inspections remain pinned to their original rule and model versions.

07

Regulatory requirements and rule model

The system must model applicability, exceptions and effective dates - not only search for keywords.

Authoritative baseline

The product baseline is the Legal Metrology Act, 2009; the official consolidated Legal Metrology (Packaged Commodities) Rules, 2011; subsequent Gazette notifications; departmental orders or SOPs; and applicable sectoral laws. Section 18 restricts manufacture, packing, sale, import, distribution and offering of non-compliant pre-packaged commodities, while Section 36 provides penalties for specified contraventions. [R1]

> **Rule versioning is mandatory**
> The PRD intentionally avoids treating one static checklist as permanent. A finding must record the inspection/event date, rule version, source, model version and officer decision so the outcome can be reproduced later.

Mandatory declaration catalogue

Declaration family	Applicability	System validation
Responsible entity identity	Name and address of manufacturer, packer, importer, brand owner or other responsible party as applicable	Presence, role, completeness, address structure, conflicts across faces/listing
Product identity	Common or generic name; component names where multiple products are packed	Presence, specificity, multi-commodity handling and consistency with product metadata
Net quantity / number	Weight, measure, volume, length, area or count using the applicable unit and format	Value-unit parsing, allowed units, placement, duplicate conflicts, category-specific declarations
Retail sale price / MRP	Price in Indian currency and required wording, including treatment of taxes, subject to active rules	Currency, amount, required phrase, strike-through/current price distinction, sticker/overprint history
Unit sale price	Unit price when required by the active rule and product context	Calculation basis, unit, rounding, relationship with quantity and MRP
Month and year / dates	Manufacture, packing or import date and best-before/use-by/expiry where applicable	Date type, format, chronology, visibility and category-specific exceptions
Consumer care	Name/address and telephone/email or other prescribed details	Presence, syntactic validation, role, missing channel and e-commerce consistency
Country of origin	Country of origin/manufacture for imported products and channel-specific display/filter duties	Import flag, country normalization, listing visibility, filter/search behavior
Dimensions / size	Dimensions where the commodity is sold by length, width, number or area	Presence, unit, value structure and product applicability
Combination / group / multipiece information	Declarations for packages containing multiple units or different commodities	Package taxonomy, quantity hierarchy, component mapping and duplication

Declaration family	Applicability	System validation
Category-specific declaration	Additional or displaced obligations under sectoral law, Gazette notification, SOP or Schedule	Approved overlay, precedence, exception, specialist review and legal citation

Presentation, placement and readability model

- Identify the likely principal display panel and all visible package faces, while allowing an officer to correct the geometry.
- Apply the exact Rule 7 table and associated provisions from the active legal source as versioned parameters, including distinct handling for blown, formed, molded, embossed or perforated declarations. [R2]
- Check character-height and width rules only when the image contains a valid physical scale, calibrated capture, known dimension or approved device-based measurement.
- Evaluate contrast, blur, glare, occlusion, curvature, compression, background clutter and text segmentation separately from legal font-size compliance.
- Detect declarations hidden by folds, stickers, seams, caps or marketplace UI truncation, and preserve a visual overlay showing the evidence region.
- Use Not Assessable where physical size, complete package coverage or legal applicability cannot be established reliably.

Current rule-routing examples included in the baseline

Context	Rule-routing requirement	Product behavior	Ref.
Medical devices	The 2025 amendment routes declaration location and font dimensions for medical-device packages to the Medical Devices Rules, 2017 while retaining the declaration obligation.	Category classifier + specialist overlay; do not apply ordinary location/font logic blindly.	[R4]
Pan masala	From 1 February 2026, pan masala packs, including small packs, must display retail sale price and mandatory declarations.	No small-pack bypass; category-specific regression tests.	[R5]
Imported products on e-commerce	From 1 July 2026, e-commerce entities selling imported products must provide a searchable and sortable country-of-origin filter.	Listing and marketplace-experience test in addition to product-page extraction.	[R6]
AEO bonded-warehouse declarations	The May 2026 amendment permits specified AEO Tier-2/Tier-3 importers to make declarations in bonded warehouses, provided retail packages are compliant before leaving.	Capture importer/AEO context and point-in-process; validate retail package at release boundary.	[R7]
Registration administration	The May 2026 amendment changes Rule 27 administration, including responsible-director and annual-update concepts.	Maintain responsible-entity registration metadata and expiry/cancellation status where integrated.	[R7]
Improvement notice workflow	Legal Metrology provisions under Jan Vishwas reforms became effective on 1 May 2026 and enable improvement notices for specified first procedural/regulatory non-compliances.	Policy-controlled notice, cure, re-inspection and escalation path; repeat/fraud exclusions handled by authorized officer.	[R8]
Edible oils	A June 2026 SOP amendment prescribes/recommends standard pack sizes for major edible oils and reinforces equivalent weight when quantity is declared by volume, with a transition period.	Category overlay with effective/transition date, package-size table and dual volume/weight declaration.	[R9]

Legal rule object

Rule-object area	Required fields
Identity	Rule ID, title, source ID, source hash, citation, jurisdiction, owner

Rule-object area	Required fields
Temporal	Publication date, effective-from, effective-to, transition period, retrospective flag
Applicability	Commodity/category, package type, package size/area, import status, channel, actor, consumer class, exemptions
Inputs	Required product attributes, extracted declaration types, package geometry, scale/calibration and external data
Logic	Presence, format, arithmetic, comparison, chronology, placement, readability, threshold and cross-field rules
Outcome	Pass/fail/review/not-applicable/not-assessable, severity, reason, remediation and report wording
Governance	Draft/approved/published/retired status, author, approver, test pack, change diff and rollback target

08

Functional requirements

MoSCoW priorities: Must = MVP launch requirement; Should = planned immediately after MVP; Could = valuable but deferrable.

8.1 Identity, access and organizational control

ID	Requirement	Priority	Acceptance signal
IAM-01	Support government/enterprise SSO through OIDC or SAML, plus controlled local accounts where approved; enforce MFA for privileged and finalizing roles.	**Must**	Authorized user can authenticate, MFA is enforced by policy and failed events are audited.
IAM-02	Provide configurable role-based access for inspector, reviewer, rule administrator, analyst, system administrator and auditor.	**Must**	Each API and UI action is denied unless the assigned permission explicitly allows it.
IAM-03	Scope data access by organization, state/UT, district, office, team, assignment and case sensitivity.	**Must**	A user cannot search, export or open records outside the effective scope.
IAM-04	Support user invitation, approval, activation, transfer, suspension, offboarding and periodic access review.	**Must**	Lifecycle events take effect within the configured SLA and remain in the audit log.
IAM-05	Implement session timeout, concurrent-session limits, device/session revocation and risk-based re-authentication.	**Must**	Policy tests pass for inactivity, privileged actions and revoked devices.
IAM-06	Enforce segregation of duties for rule publication, final report approval and audit-log administration.	**Must**	The same actor cannot complete prohibited maker-checker combinations.
IAM-07	Record immutable authentication, authorization and administrative audit events.	**Must**	Audit query shows actor, timestamp, source, action, object, outcome and correlation ID.

8.2 Inspection creation, intake and field capture

ID	Requirement	Priority	Acceptance signal
INS-01	Create an inspection with auto-generated ID, jurisdiction, location, channel, date/time, officer, seller/entity and product context.	**Must**	Required fields and permitted defaults are validated before submission.
INS-02	Support physical retail, warehouse/import, back-office, complaint-	**Must**	Type selection changes the required

ID	Requirement	Priority	Acceptance signal
	driven and e-commerce inspection types.		metadata and workflow without data loss.
INS-03	Accept camera capture and JPEG, PNG, HEIC and PDF upload, with configurable size/page limits and malware scanning.	Must	Allowed files ingest; disallowed or unsafe files fail with a clear reason.
INS-04	Capture multiple labeled package views and close-ups; show a coverage checklist by package form.	Must	Inspection cannot be auto-evaluated as complete until mandatory views are supplied or waived with reason.
INS-05	Provide live camera guidance for distance, focus, glare, orientation, edge coverage and calibration-marker placement.	Should	User receives actionable recapture prompts before leaving capture mode.
INS-06	Capture timestamp, device, optional GPS, source channel and uploader; allow policy-based location correction with audit.	Must	Metadata is preserved and edits create before/after audit entries.
INS-07	Capture or scan barcode/GTIN, SKU, brand, generic name, category, batch/lot and package size where available.	Must	Identifiers can be searched and used for duplicate/product-history matching.
INS-08	Support bulk intake through drag-and-drop, manifest spreadsheet or API for back-office campaigns.	Should	System validates each row/file and produces a row-level error report.
INS-09	Support URL, full-page screenshot, PDF print or structured feed as e-commerce evidence.	Must	Source and evidence are timestamped and linked to the inspection.
INS-10	Detect potential duplicates and allow re-inspection or linked-case creation without overwriting prior evidence.	Must	Potential matches are presented; new inspection receives a unique ID and lineage.
INS-11	Autosave drafts, support assignment/reassignment, due dates, status and prioritized queues.	Must	No accepted edit is lost after refresh or connectivity interruption.
INS-12	Allow configurable checklists and campaign templates by category, jurisdiction and enforcement objective.	Should	Admin can publish a template and new inspections inherit its fields and view requirements.

8.3 Image processing, OCR and declaration extraction

ID	Requirement	Priority	Acceptance signal
VIS-01	Preserve original evidence as immutable content, compute a cryptographic hash and store all derivatives as separate versions.	Must	Original hash remains stable; edits never replace the original binary.
VIS-02	Perform rotation correction, perspective dewarping, crop assistance, denoise and contrast normalization while retaining the original.	Must	Officer can compare original and derivative and revert processing for review.
VIS-03	Score blur, glare, occlusion, resolution, text scale, curvature, clipping and package-face completeness.	Must	Low-quality inputs trigger a specific reason and recapture recommendation.
VIS-04	Detect package surfaces and assist identification of the principal display panel and declaration clusters.	Must	Detected geometry is editable and changes re-run affected checks.
VIS-05	Accept a calibration marker, known physical dimension or approved device measurement for millimetre estimation.	Must	Calibration method, value and uncertainty are stored with the result.
EXT-01	Recognize English and Hindi at MVP, with architecture and training path for additional Indian scripts and mixed-script labels.	Must	Benchmark results are reported by script, package type and image-quality band.
EXT-02	Return text, word/line bounding boxes, reading order, script, confidence and source image for every extracted block.	Must	Clicking extracted text highlights the exact evidence region.
EXT-03	Classify declaration candidates such as MRP, quantity, dates, responsible entity, consumer care, country of origin and unit price.	Must	Each candidate has type, value, confidence and provenance.

ID	Requirement	Priority	Acceptance signal
EXT-04	Normalize currency, units, numbers, dates, phone/email and country names while preserving verbatim text.	Must	Normalized value is reversible to source text and ambiguous parses require review.
EXT-05	Merge declaration evidence across faces and detect duplicates, conflicts or continuation text.	Must	System presents a unified declaration with all contributing image regions.
EXT-06	Allow officers to correct text, type, value, language, bounding box and declaration grouping.	Must	Correction records original value, new value, actor, reason and timestamp.
EXT-07	Expose confidence and extraction limitations without converting confidence into a legal compliance score.	Must	UI distinguishes model confidence from legal status.

8.4 Product classification and rule-based compliance engine

ID	Requirement	Priority	Acceptance signal
REG-01	Store rules, thresholds, sources and report wording as versioned configuration with effective-from/effective-to dates.	Must	Historical inspection can reproduce the exact applied rule version.
REG-02	Select rules using product category, package type, package dimensions/PDP area, import status, sales channel, actor, consumer class, jurisdiction and event date.	Must	Boundary and exception test pack selects the expected rule set.
REG-03	Evaluate presence and completeness of all mandatory declarations applicable to the selected context.	Must	Each requirement returns a status, reason, evidence and citation.
REG-04	Validate syntactic and semantic format, allowed unit/currency/date patterns and role-specific address/contact requirements.	Must	Invalid and ambiguous values are separately identified.
REG-05	Detect conflicting values across package faces, stickers, listing fields and physical-vs-online evidence.	Must	Conflict finding names both values and links both evidence regions.
REG-06	Test net quantity/number declarations, units and category-specific quantity relationships.	Must	Approved positive, negative and edge cases pass regression tests.
REG-07	Test MRP/RSP wording, Indian-currency amount, tax-inclusive treatment and unit-sale-price rules according to the active version.	Must	Rule outcome includes parsed values and calculation evidence where applicable.
REG-08	Validate manufacture/packing/import/best-before/use-by/expiry date type, format and chronology where applicable.	Must	Impossible or contradictory chronology is flagged with field-level explanation.
REG-09	Validate responsible-entity identity/address, import status and country-of-origin obligations.	Must	Missing role or mismatch is detected without assuming every role is always required.
REG-10	Validate consumer-care fields and the applicable contact channels.	Must	Missing/invalid contact information receives a distinct finding.
REG-11	Support combination, group and multipiece package logic, including parent/child quantities and component names.	Should	Test packages resolve to the correct hierarchy and declarations.
REG-12	Model exemptions, sectoral overlays, transitional provisions and legal precedence; never infer universal applicability.	Must	Officer can see why a rule was included, excluded or superseded.
REG-13	Produce Passed, Failed, Needs Review, Not Applicable and Not Assessable outcomes with severity and remediation guidance.	Must	No forced binary result occurs when evidence or applicability is insufficient.
REG-14	Attach a human-readable legal explanation and source citation to every rule evaluation.	Must	Report and UI cite the exact approved source/version used.
REG-15	Support historical replay, rule simulation, comparison between versions and impact analysis before publication.	Should	Admin can run the same test pack under two versions and see outcome diffs.

ID	Requirement	Priority	Acceptance signal
REG-16	Keep deterministic legal rules separate from probabilistic model outputs and log each component version.	Must	A model deployment cannot silently alter statutory logic.

8.5 Font size, placement and readability

ID	Requirement	Priority	Acceptance signal
FNT-01	Estimate physical character height only when valid scale/calibration or approved measurement metadata is available.	Must	Without calibration, physical-size status is Not Assessable.
FNT-02	Implement the active Rule 7 table and package-form variants as configurable, source-linked data.	Must	Every threshold row has source, effective dates and boundary tests.
FNT-03	Measure numeral/letter height, approximate width-to-height ratio and line spacing at declaration-region level.	Must	Measurement overlay displays sample characters, scale and uncertainty.
FNT-04	Treat molded/embossed/perforated declarations as a distinct presentation mode with its own active thresholds.	Must	Package-form classification is visible and editable before evaluation.
FNT-05	Assess readability using contrast, blur, glare, occlusion, curvature, clutter, compression and image resolution.	Must	Readability reason codes are separate from font-size reason codes.
FNT-06	Check placement on/relative to the principal display panel, grouping, visibility and obstruction where legally applicable.	Must	Finding highlights expected context and actual evidence location.
FNT-07	Allow manual measurement entry from an approved physical method and preserve the instrument/method metadata.	Should	Officer-entered measurement is distinguished from computer-vision estimate.
FNT-08	Provide evidence overlays and downloadable annotated images for each presentation finding.	Must	Report contains the same region and measurement seen during review.

8.6 E-commerce and digital-listing compliance

ID	Requirement	Priority	Acceptance signal
ECM-01	Preserve URL, marketplace, seller, timestamp, screenshot/PDF and available structured page data.	Must	Source package can be independently opened or verified from retained evidence.
ECM-02	Extract required listing declarations from visible page content and distinguish product-page, cart and seller-panel evidence.	Must	Each extracted field is linked to its page region or structured source.
ECM-03	Validate responsible-entity, imported-product country-of-origin, consumer-care, MRP and net-quantity information according to the active channel rule.	Must	Missing listing fields generate channel-specific findings.
ECM-04	Test searchable/sortable country-of-origin filter behavior for imported products where applicable from 1 July 2026.	Should	Test records query, selected value, result behavior, timestamp and screenshot.
ECM-05	Compare online declarations with physical-pack declarations and show additions, omissions and contradictions.	Should	Difference view links both evidence sources and preserves versions.
ECM-06	Handle dynamic/lazy-loaded content, regional variants and login-gated evidence through an officer-assisted capture flow.	Should	Manual fallback produces a complete timestamped evidence package.
ECM-07	Respect authorized access, marketplace agreements and configured crawl limits; provide no-crawl/manual paths.	Must	System honors domain policy, rate limit and robots/access configuration.

8.7 Review, evidence and enforcement case workflow

ID	Requirement	Priority	Acceptance signal
REV-01	Provide queues by assignment, severity, confidence, SLA, category, campaign and jurisdiction.	Must	Authorized user can filter, sort and bulk assign permitted records.
REV-02	Allow finding disposition: confirm, dismiss, edit, merge, split, defer, not applicable or not assessable, with reason.	Must	Every disposition is versioned and attributable.
REV-03	Require an override reason and, for configured high-risk changes, second-level approval.	Must	Finalization is blocked until required approval is complete.
REV-04	Support image annotation, crop, callout, measurement line, comment and supporting-document attachment.	Must	Annotations are non-destructive and included in the evidence manifest.
REV-05	Support configurable severity, legal provision, recommended remedy, due date and escalation path.	Must	Severity and workflow are controlled by approved policy, not model confidence alone.
REV-06	Support warning/advisory, improvement notice, re-inspection, closure and escalation workflows where legally/policy applicable.	Should	Case timeline includes service, response, cure evidence and outcome.
REV-07	Link related products, entities, prior inspections, complaints, notices and repeat findings.	Must	User can navigate lineage without duplicating or overwriting records.
REV-08	Lock finalized reports/findings; amendments create a new signed version with reason.	Must	Prior final version remains accessible and hash-verifiable.

8.8 Reports, repository, search and export

ID	Requirement	Priority	Acceptance signal
RPT-01	Generate a standardized compliance/non-compliance report with inspection, product, rule, finding, evidence, officer and audit sections.	Must	Template passes approved sample-content and pagination tests.
RPT-02	Export signed/final PDF and editable DOCX or ODT; offer CSV/JSON for structured data and evidence manifest.	Must	Exports contain the same approved data and version identifiers.
RPT-03	Embed photographs, bounding boxes, annotations, extracted text, confidence and measurement details as configured.	Must	Every material finding has at least one evidence reference or an explicit no-evidence reason.
RPT-04	Include legal citations, active rule version, report version, model version and officer disposition.	Must	A reviewer can reproduce the applied logic from report metadata.
RPT-05	Support digital signature/e-sign integration, QR verification and cryptographic hash of the report package.	Should	Verifier can confirm authenticity and retrieve the permitted verification record.
RPT-06	Support English, Hindi and bilingual templates, plus configurable agency branding.	Should	Same case renders correctly in each approved template.
REP-01	Maintain a repository of products, inspections, evidence, findings, reports, notices, actions and audit history.	Must	Records remain linked and searchable through lifecycle states.
REP-02	Provide full-text OCR search and filters for identifier, brand, commodity, entity, location, date, rule, status and severity.	Must	Typical query returns within the performance target and honors data scope.
REP-03	Show a product/entity timeline, duplicate suggestions, repeat findings and related-case graph.	Should	Timeline is ordered, source-linked and exportable.
REP-04	Support configurable retention, legal hold, archive, restore and policy-controlled disposal.	Must	Disposal requires authorization and leaves a policy audit record.

8.9 Dashboards, administration and integrations

ID	Requirement	Priority	Acceptance signal
DASH-01	Provide executive and operational dashboards for inspections, compliance status, violations, workload and SLA.	Must	Metrics are filterable, drillable and reconcile with repository records.
DASH-02	Analyze violations by declaration, rule, category, brand, responsible entity, marketplace, geography and time.	Must	Charts support export and preserve applied filters.
DASH-03	Highlight repeat entities/products, emerging patterns, data-quality gaps and high-risk clusters.	Should	Pattern logic is explainable and does not automatically establish liability.
DASH-04	Provide model-quality dashboard by script, package type, category, confidence band and review outcome.	Must	Domain owner can see drift and false-negative indicators.
ADM-01	Provide no-code/low-code legal rule administration with source attachments, effective dates and maker-checker publication.	Must	Approved admin can draft, test, compare, publish and roll back a rule version.
ADM-02	Manage commodity taxonomy, declaration synonyms, units, countries, jurisdictions, severity and templates.	Must	Configuration changes are versioned and testable.
ADM-03	Maintain model registry, deployment status, benchmark results and rollback.	Must	Every inference links to an approved model version.
MOB-01	Provide responsive web/PWA and an Android-first field experience with encrypted offline drafts.	Must	Core capture/review works offline and syncs later without duplicate records.
MOB-02	Resolve sync conflicts, show pending/failed uploads and support policy-controlled remote sign-out/wipe.	Must	Conflict outcome is explicit and audited.
INT-01	Expose authenticated REST APIs for inspection, assets, processing, findings, review, reports, search, rules and audit events.	Must	APIs are versioned, rate-limited, documented and support idempotency.
INT-02	Support webhooks/events for processing completion, assignment, due date, report finalization and rule publication.	Should	Delivery retries and dead-letter handling are observable.
INT-03	Integrate through adapters with SSO, e-sign, messaging, barcode/GTIN data, case systems, marketplaces and data warehouse tools as approved.	Should	Adapter failure does not corrupt the inspection and has a manual fallback.

09

AI, OCR and computer vision requirements

Model responsibilities

Component	Purpose	Output contract
Image quality model	Blur, glare, occlusion, clipping, perspective, resolution and capture guidance	Reason codes + score; never a legal conclusion
Package/layout model	Package faces, likely PDP, text regions, declaration clusters, stickers/overprints	Editable geometry + confidence
OCR/script model	Verbatim text, word/line boxes, reading order, language/script	Text + boxes + confidence + model version
Declaration classifier / NER	Declaration type, entity/value boundaries and field normalization	Candidate fields + evidence links
Font measurement pipeline	Character segmentation and physical-size estimate using calibration	Measurement + uncertainty + method; otherwise Not Assessable
Anomaly assistance	Contradictory values, unexpected wording, duplicate listing or recurring pattern	Needs Review suggestion, not automatic liability

Human-in-the-loop safeguards

- No enforcement report is finalized solely from AI output; an authorized officer must review material findings.
- The UI displays source crop, verbatim OCR, normalized value, confidence, model version and rule explanation together.
- Low-confidence extraction, conflicting faces, missing scale and unfamiliar categories automatically route to Needs Review or Not Assessable.
- Officer corrections are stored as adjudicated labels for governed model evaluation; they are not used for training without approval and data-governance controls.
- A model change cannot alter published legal rules, previously finalized reports or historical evidence.

Evaluation and MLOps

Area	Requirement
Benchmark corpus	Representative brands/categories, English/Hindi/mixed script, curved/glossy/molded packs, small type, e-commerce screenshots, positive/negative/boundary cases
Data splits	No near-duplicate SKU or image leakage across train, validation and legal acceptance test sets
Metrics	OCR CER/WER, declaration precision/recall/F1, field exact/normalized match, calibration error, font MAE/pass-fail, quality-gate recall, review override rate
Release gates	Target metrics met by major segment; critical false negatives manually reviewed; security/privacy/legal approvals complete
Monitoring	Input drift, confidence shift, script/category mix, latency, failure rate, officer corrections, false-negative sampling and version comparison
Rollback	One-click rollback to last approved model; queued jobs preserve requested version and idempotency

Critical metric

Recall for missing high-risk declarations is more important than a visually impressive overall accuracy score. Acceptance reporting must be segmented by declaration, script, category, package type and image-quality band.

10

User experience, accessibility and localization

Information architecture

Area	Experience requirement
Home / dashboard	Assigned work, due items, inspection summary, trends and quick-create action
New inspection	Context, product identifiers, package/listing evidence and guided capture
Analysis workbench	Image viewer, OCR/declarations, rule findings, evidence overlay and officer decision in one workspace
Case timeline	Inspection, notice, response, re-inspection, report versions and linked history
Repository	Full-text search, faceted filters, saved views, product/entity timelines and exports
Rule library	Published versions, sources, effective dates and human-readable changes; admin editing is permission-controlled
Administration	Users, roles, taxonomy, integrations, templates, retention and system health

Interaction standards

- Desktop workbench optimizes side-by-side evidence and findings; mobile prioritizes capture and quick review.
- Statuses use text, icon and color together: Compliant, Non-compliant, Needs Review, Not Applicable and Not Assessable.
- Every finding has a plain-language explanation, legal citation, evidence crop and available corrective action.
- Destructive or final actions require confirmation, permission and reason; autosave and undo are used for draft actions.
- System meets WCAG 2.2 AA target, including keyboard navigation, focus order, contrast, labels, alt text and screen-reader semantics.
- Initial UI and report languages are English and Hindi; text architecture supports Unicode, mixed scripts and locale-specific numbers/dates.

Landing-page design direction

The public or sign-in landing experience should communicate three promises immediately: scan the label, verify the applicable declarations, and preserve evidence. The concept in Appendix A uses a trustworthy government-enterprise visual language, clear primary action, a realistic inspection workbench preview and feature proof points rather than generic AI claims.

11

Data model, records and retention

Core entities

Entity	Key content
Organization / jurisdiction	Agency hierarchy, office, geography, scope and policy
User / role / assignment	Identity, permissions, team, delegation and workload
Inspection	Type, context, location, dates, status, event date, rule/model pinning
Product / SKU / package	Brand, generic name, category, identifiers, package form, size and import status
Entity	Manufacturer, packer, importer, brand owner, seller, marketplace and consumer-care entity
Evidence asset	Original image/page/file, hash, metadata, derivatives, annotations and chain of custody
OCR block / declaration	Verbatim text, region, script, normalized value, confidence, correction history
Rule set / rule version	Source, applicability, logic, outcomes, dates, test pack and approvals
Evaluation / finding	Rule result, severity, reason, evidence, confidence, officer disposition and citation
Report / notice / action	Template, version, signature, service, due date, response, re-inspection and closure
Audit event	Actor/service, event, object, before/after, source, timestamp and correlation ID
Model version	Artifact, configuration, benchmark, approval, deployment and rollback state

Evidence integrity rules

- Original evidence is write-once; every derivative references the original hash and processing operation.
- Finalized findings and reports are immutable; correction creates a new version and retains superseded content.
- Every record carries tenant/jurisdiction scope, created/updated actor and timestamps in synchronized time.
- Evidence export includes a manifest of files, hashes, versions, relationships and signature/verification data.
- Retention duration is policy-configurable by record type and jurisdiction; legal hold blocks deletion and archive disposal.

Search and indexing

The search index will include permitted product metadata, responsible entities, OCR text, normalized declarations, findings and report identifiers. Search results must enforce the same data-scope authorization as the transactional system. The repository must support exact, fuzzy and faceted search, while legal rule and audit records remain retrievable by version and source.

12

Solution architecture and deployment

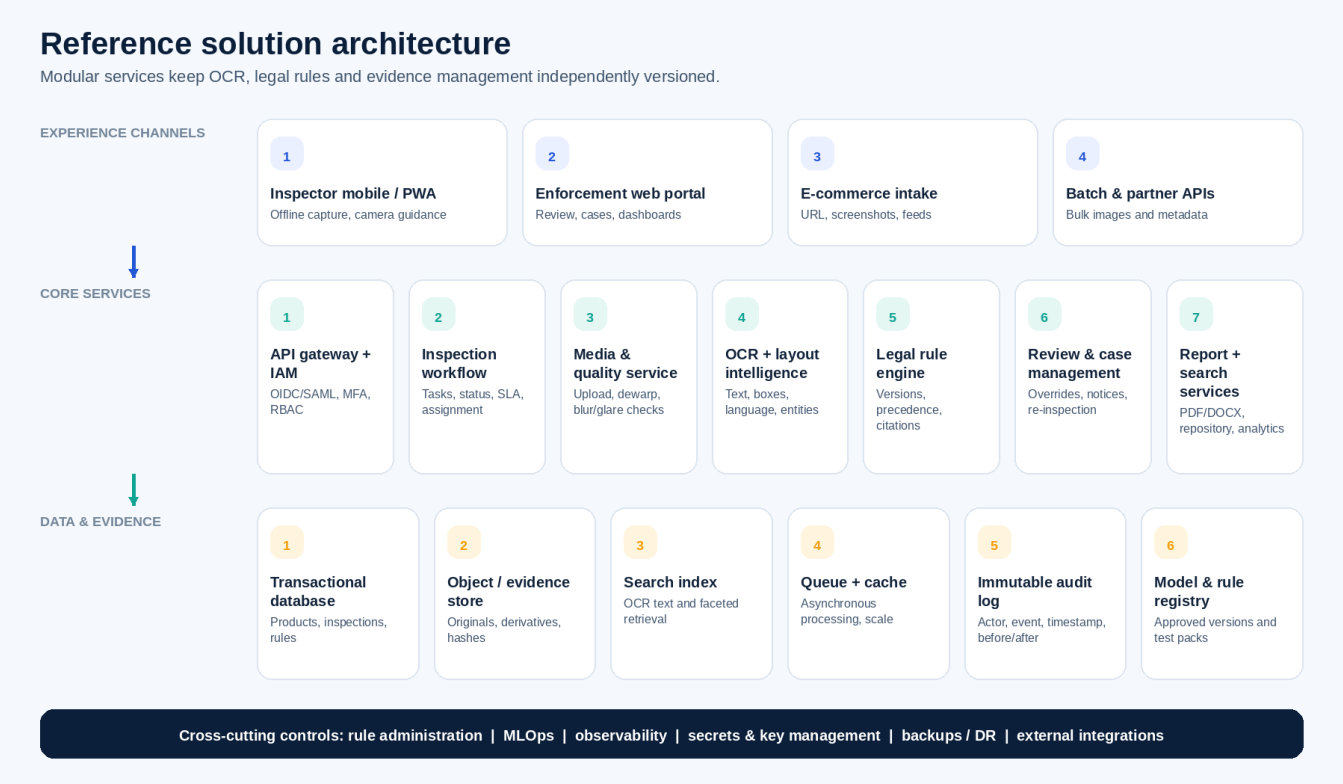


Figure 2. Logical architecture. Physical deployment may begin as a modular application with independently scalable ML workers.

Architecture principles

- Separate evidence ingestion, model inference, deterministic rules and officer workflow so each can be versioned and scaled independently.
- Use asynchronous processing for image/OCR jobs with idempotent APIs, retries, dead-letter handling and visible job status.
- Pin every evaluation to evidence, rule and model versions; do not recompute finalized results silently.
- Store transactional state in a relational database, binary evidence in secure object storage, and searchable text in a dedicated index.
- Expose integration adapters at the boundary so marketplace, SSO, e-sign and case systems can change without modifying core logic.
- Support government cloud, approved private cloud or on-premises deployment through containerization and infrastructure-as-code.

Recommended technology-neutral components

Layer	Capability
Frontend	Responsive web/PWA; Android shell/native capture if device APIs or offline controls require it

Layer	Capability
API and workflow	REST gateway, IAM adapter, orchestration/workflow service and notification service
Image/ML	GPU/CPU worker pool, image preprocessing, OCR/layout/NER and model-serving layer
Rules	Versioned deterministic rule engine, rule registry, simulation and legal test runner
Data	Relational database, object/evidence store, search index, cache and message queue
Reporting	Template engine, PDF and DOCX/ODT generation, signature/e-sign and verification endpoint
Operations	Central logging, metrics, tracing, SIEM feed, secrets/KMS, backup, DR and configuration management

Deployment environments

Environment	Purpose
Development	Synthetic/de-identified data; rapid model/rule iteration; no production credentials
Integration / QA	System integration, migration, performance, security and automated regression tests
Legal UAT	Approved rule sources and locked legal test pack; controlled representative evidence
Staging	Production-like sizing, network, IAM, backups, monitoring and release rehearsal
Production	Isolated access, approved data residency, HA, monitoring, backups, DR and change management

API surface

API domain	Core operations
Inspection API	Create/update/assign/status, product/entity context and lineage
Asset API	Initiate upload, integrity check, metadata, derivative status and signed retrieval
Processing API	Submit job, status, model version, results and retry/cancel
Evaluation API	Rule selection, evaluations, explanation, simulation and version comparison
Review API	Correction, disposition, annotation, override, approval and finalization
Report API	Generate, sign, download, verify and export manifest
Search/analytics API	Scoped search, aggregations, saved views and scheduled extract
Administration API	Users/roles, taxonomy, rules, models, templates, retention and integrations

13

Security, privacy and evidence integrity

Security requirements

Domain	Minimum control
Identity	SSO, MFA, least privilege, maker-checker, privileged access management and periodic recertification
Data protection	TLS in transit, strong encryption at rest, managed keys, secret vault and protected backups

Domain	Minimum control
Application	Secure SDLC, code review, dependency/container scanning, SAST/DAST, API validation, rate limiting and CSRF/XSS/SQLi controls
Evidence	Immutable originals, hashes, signed manifests, versioned annotations, synchronized time and chain-of-custody events
Audit	Append-only security and business audit, SIEM integration, alerting, protected retention and independent access
Infrastructure	Network segmentation, WAF, hardened images, patching, vulnerability management, endpoint/device controls and environment isolation
Privacy	Data minimization, purpose limitation, masking, controlled export, retention/legal hold and approved handling of personal/contact data
Incident response	Detection, triage, containment, evidence preservation, notification and tested recovery procedures

Evidence and report authenticity

- Hash every original asset at intake and include asset hashes in the evidence manifest.
- Hash and optionally digitally sign each finalized report package; display verification status and report version.
- Use a QR or verification code that discloses only approved public verification data, not sensitive case content.
- Record all reads, downloads, exports and administrative access to sensitive evidence where policy requires.
- Use synchronized, monitored time sources; preserve source-device time separately from server-received time.

AI data boundary

Production evidence must not be sent to unapproved public AI services. Model hosting, telemetry and support access must conform to the approved deployment and data-residency architecture.

14

Non-functional requirements

ID	Quality attribute	Acceptance target
NFR-PERF-01	First useful analysis for up to five standard images	P95 <=60 seconds under agreed pilot load; progress visible within 3 seconds
NFR-PERF-02	Batch processing	Queue and process at configured concurrency without blocking interactive users; row/job status available
NFR-SEARCH-01	Typical repository search	P95 <=2 seconds for agreed indexed query set
NFR-AVAIL-01	Production availability	>=99.5% monthly for MVP, excluding approved maintenance
NFR-SCALE-01	Horizontal scale	Stateless services and worker pools scale independently; capacity test target agreed before pilot
NFR-RELIAB-01	Job reliability	Idempotent processing; no duplicate final records; retry and dead-letter recovery tested
NFR-DR-01	Disaster recovery	Target RPO <=15 minutes and RTO <=4 hours, subject to hosting approval and cost
NFR-OFFLINE-01	Field resilience	Create/capture/edit draft offline; encrypted sync resumes safely after connectivity returns
NFR-SEC-01	Security baseline	Zero unresolved critical vulnerabilities at release; high findings accepted only by authority
NFR-AUD-01	Auditability	100% of final findings and privileged changes attributable and version-reproducible

ID	Quality attribute	Acceptance target
NFR-ACC-01	Accessibility	WCAG 2.2 AA target for web workbench and keyboard-only critical flows
NFR-LOC-01	Localization	Unicode end-to-end; English/Hindi MVP; locale-aware dates/numbers and mixed-script evidence
NFR-MAINT-01	Maintainability	Rules, templates, taxonomy and thresholds update without application redeployment
NFR-OBS-01	Observability	Central logs, metrics, traces, correlation IDs, model/rule telemetry and actionable alerts
NFR-PORT-01	Portability	Containerized deployment, documented APIs and export of reports/evidence/structured data in open formats

Capacity assumptions to validate during discovery

Assumption	Planning value	Why it matters
Pilot jurisdictions	To be confirmed	Drives tenants, data scope and support model
Daily inspections	To be confirmed by campaign and state	Drives OCR worker and storage sizing
Images per inspection	Default planning range: 5-12	Drives upload, latency and evidence storage
Retention	Policy-defined by record type	Drives object-store lifecycle and archive design
Concurrent users	Pilot benchmark to be agreed	Drives API, database, search and session sizing
Language/category mix	Measure in discovery corpus	Drives benchmark and model prioritization

15

Reports, dashboards and analytics

Compliance report content

Report section	Required content
1. Identification	Report ID/version, inspection ID, jurisdiction, dates, location/channel, officer and approval
2. Product and parties	Product/SKU, brand, category, package form/size, barcode, seller and responsible entities
3. Evidence manifest	Original files, thumbnails, source URL, timestamps, device/source, hashes and annotations
4. Applied legal context	Event date, rule set/version, category overlay, exemptions considered and model versions
5. Declaration results	Expected declaration, extracted text/value, status, confidence, evidence region and officer disposition
6. Violations / observations	Severity, explanation, exact citation, evidence, recommended action and corrective requirement
7. Officer conclusion	Final status, notes, override reasons, approval and signature/e-sign
8. Audit and verification	Report hash/QR, generation time, version history and evidence package reference

Dashboard views

View	Minimum metrics
Enforcement overview	Inspections, compliant/non-compliant/review status, high-severity findings, trend and geography
Operations	Assigned/pending/overdue, processing failures, review backlog, SLA and officer/team workload
Violation intelligence	Top rules/declarations, categories, responsible entities, marketplaces, repeat products and time trends
Corrective action	Notices due, responses, cured/not cured, re-inspections and escalation
Data/model quality	Image recapture rate, OCR confidence, corrections, override rate, false-negative audits and drift
Rule administration	Upcoming effective dates, unpublished drafts, failed legal tests, version impact and source freshness

Analytics guardrail

Risk indicators prioritize inspection resources; they must not be shown as proof of a legal contravention. Every aggregate should permit drill-down to the underlying reviewed records and applied filters.

16

Testing, acceptance and release gates

Acceptance scenarios

ID	Scenario	Expected outcome
UAT-01	Complete physical inspection	User captures all faces, quality gate passes, declarations extract, findings review and final report export succeeds.
UAT-02	Missing core declaration	Approved sample lacking a required declaration produces the expected failed finding, citation and evidence context.
UAT-03	Calibrated font failure	Known scale and below-threshold text produce expected measurement, uncertainty, finding and overlay.
UAT-04	No scale available	Same image without scale returns Not Assessable for physical font size, not pass or fail.
UAT-05	Historical rule replay	An inspection before/after an effective date selects different approved versions and preserves each result.
UAT-06	E-commerce imported product	Listing extraction and country-of-origin/filter test follow the active channel rule and preserve timestamped evidence.
UAT-07	Special category routing	Medical device, pan masala and edible-oil samples invoke the correct overlay or specialist review path.
UAT-08	Officer correction and override	Extraction correction and finding override require reason and appear in audit/report history.
UAT-09	Improvement notice lifecycle	Eligible policy scenario creates notice, due date, response, re-inspection and closure/escalation lineage.
UAT-10	Access-control isolation	User cannot view/search/export out-of-scope inspections, even through direct API or report URL.
UAT-11	Offline sync	Field draft and images created offline sync once, preserve order and recover cleanly from interrupted upload.
UAT-12	Evidence verification	Report/evidence manifest hashes verify; tampered file is detected.

Test layers

Layer	Minimum coverage
Unit and component	Parsers, normalizers, rule operators, role checks, report fragments and hash utilities
Rule regression	Positive/negative/boundary/exemption/transition/historical cases for every published rule
Model evaluation	Locked corpus segmented by field, script, category, package form and quality band
Integration	IAM, storage, search, OCR, e-sign, messaging, marketplace and case adapters
End-to-end / UAT	Representative field and e-commerce journeys with authorized officers
Performance / resilience	Load, soak, queue recovery, retries, offline sync, backup restore and DR exercise
Security / privacy	Threat model, SAST/DAST, penetration test, access isolation, export controls and incident runbook
Accessibility	Automated and manual keyboard/screen-reader checks for critical flows

Release gates

- Legal: domain authority approves active rule sources, machine-readable logic, explanations and locked test pack.
- Product/UAT: all Must requirements accepted or formally waived; no open critical workflow defect.
- Model: target metrics met on approved corpus; critical false-negative review complete; rollback tested.
- Security: production threat model, penetration test, vulnerability remediation and access review approved.
- Operations: monitoring, backups, restore, DR, support, escalation, runbooks and training ready.
- Data: migration/seed data validated, retention configured and sample evidence verification passes.

17

Roadmap, risks and governance

Indicative delivery roadmap

Phase	Indicative duration	Deliverables
0. Discovery and legal design	4-6 weeks	Rule inventory, stakeholder interviews, corpus audit, architecture, security/data assessment, UX prototypes and pilot plan
1. MVP build	16-20 weeks	Core web/PWA, capture, quality, English/Hindi OCR, extraction, versioned rules, review, PDF/editable reports, repository, RBAC and dashboards
2. Pilot and hardening	8-12 weeks	Field pilot, benchmark expansion, rule/model tuning, performance/security testing, training, integrations and production readiness
3. Expansion	12-16 weeks	Additional scripts, e-commerce automation, corrective-action workflow, category overlays, advanced analytics and partner APIs
4. National-scale intelligence	Ongoing	Risk-based sampling, broader integrations, predictive surveillance, cross-jurisdiction insights and continuous rule/model governance

Durations are planning assumptions, not commitments. Final estimates depend on the legal rule inventory, image corpus quality, languages/categories, hosting accreditation, integrations and pilot scale.

Key risks and mitigations

Risk	Impact	Mitigation
Legal change or interpretation	Static logic becomes wrong; inconsistent enforcement	Versioned sources/effective dates, dual approval, regression packs, legal change monitoring and emergency rollback
Poor image quality / missing faces	False missing declaration or unreadable result	Capture guidance, quality gate, required-view checklist, Not Assessable state and recapture workflow
No physical scale	Unreliable font-size conclusion	Calibration marker/known dimension/manual measurement; never infer millimetres without valid scale
OCR/script/category bias	Unequal error rates and false negatives	Segmented benchmarks, human review, targeted data collection, drift monitoring and staged language rollout
Over-automation of legal judgment	Inappropriate enforcement action	Deterministic explanations, mandatory officer review, override/approval and clear legal disclaimer
Marketplace access variability	Incomplete or stale online evidence	Timestamped manual capture, partner feeds, no-crawl policy and source/evidence provenance
Evidence tampering or leakage	Loss of trust/admissibility and privacy breach	Immutable originals, hashes, signed manifests, scoped access, encryption, audit and incident response
Integration/hosting delays	Pilot slippage	Adapter architecture, mock services, manual fallback, early security/network discovery and phased integration
Vendor/model lock-in	High cost and inability to reproduce	Open APIs/formats, model/rule registry, exportable corpus/test packs and containerized deployment

Governance cadence

Forum	Minimum agenda
Weekly product/engineering	Scope, blockers, metrics, defects and decisions
Fortnightly domain review	Rule questions, false negatives, category routing and test-pack changes
Monthly model review	Performance segments, drift, correction/override trends and deployment approval
Quarterly access/security review	Privileged access, incidents, vulnerabilities, exports and retention exceptions
On every legal change	Source validation, impact analysis, rule test, dual approval, release note and effective-date activation

Open decisions before build

- Owning department/agency, tenant model and pilot states/UTs.
- Exact MVP categories, exemptions and sectoral overlays to be legally certified.
- Approved mobile device profile, calibration marker and physical-measurement procedure.
- Hosting, data-residency, security accreditation and network constraints.
- Required languages/scripts and minimum benchmark sample counts per category.
- Report/notice templates, signature authority and maker-checker policy.
- Integration priority for SSO, e-sign, case management, barcode data and e-commerce partners.
- Retention periods, legal-hold authority and public verification disclosure.
- Pilot volume, concurrency, SLA and operational support model.

18

Requirement traceability matrix

User-provided requirement	PRD coverage	Acceptance
Scan/analyze package images	INS-03 to INS-06; VIS-01 to VIS-05; EXT-01 to EXT-02	UAT-01, UAT-03, UAT-04
Detect mandatory declarations	EXT-03 to EXT-05; REG-03 to REG-10	UAT-02, UAT-07
Correctness, completeness and placement	REG-04 to REG-14; FNT-03 to FNT-06	UAT-02, UAT-03, UAT-07
Missing/non-compliant declarations	REG-03 to REG-13	UAT-02
Readability and font size	VIS-03 to VIS-05; FNT-01 to FNT-08	UAT-03, UAT-04
Compliance reports / violation summaries	RPT-01 to RPT-06	UAT-01, UAT-12
Photographs and supporting evidence	VIS-01; REV-04; RPT-03	UAT-01, UAT-12
Repository and compliance history	INS-10; REP-01 to REP-04	UAT-05, UAT-09
Role-based access and authentication	IAM-01 to IAM-07	UAT-10
Dashboards for officials	DASH-01 to DASH-04	Dashboard reconciliation test
Search and retrieval	REP-02 to REP-03	Search performance and scope test
PDF and editable export	RPT-02, RPT-05 to RPT-06	Report parity and verification test
E-commerce product information	INS-09; ECM-01 to ECM-07	UAT-06
Technical architecture/deployment documentation	Sections 11-14; INT-01 to INT-03	Architecture, security and operations gate

Definition of ready for implementation

- All Must requirements have an owner, final acceptance criterion, dependency and estimate.
- Legal baseline and MVP rule inventory are approved, including exemptions and category overlays.
- Representative corpus is licensed/approved, de-identified where required and quality-audited.
- UX prototype is tested with field inspectors and reviewers, including offline and font-calibration flows.
- Hosting/security/integration decisions and non-functional capacity targets are approved.

APPENDIX A

Landing-page concept

The visual below is a concept, not a final production design. It demonstrates the proposed brand hierarchy, primary call to action, enforcement-oriented messaging and an inspection workbench preview. Final UI will be validated with field officers and accessibility testing.

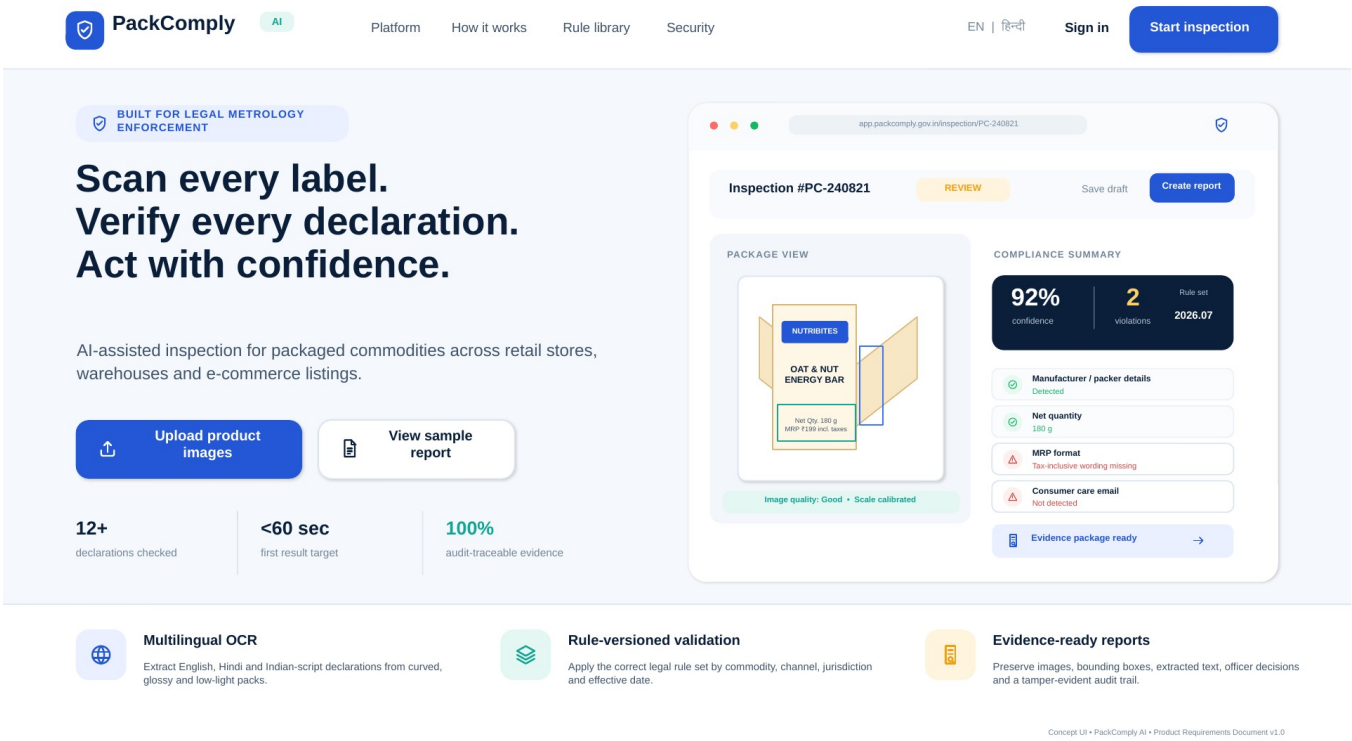


Figure 3. PackComply AI landing-page mockup. An editable PowerPoint source is provided separately.

Design system direction

Element	Direction
Tone	Trustworthy, evidence-led and operational; avoid consumer-tech hype or unsupported AI promises
Color	Navy for authority, blue for primary actions, teal for verified/assistive states, amber for review, red for violations
Typography	High-legibility sans serif with clear hierarchy and comfortable table density
Components	Guided capture, evidence viewer, declaration cards, legal explanation, status chips and report actions
Accessibility	Text/icon/color status redundancy, high contrast, keyboard paths and responsive layouts

APPENDIX B

Glossary and official references

Glossary

Term	Meaning
AEO	Authorized Economic Operator

Term	Meaning
AI	Artificial intelligence
API	Application programming interface
CER/WER	Character error rate / word error rate
E-commerce entity	Digital channel actor as defined by the applicable active legal rule
Evidence manifest	Machine-readable list of retained files, hashes, versions and relationships
LMPC Rules	Legal Metrology (Packaged Commodities) Rules, 2011, as amended
MRP/RSP	Maximum/retail sale price terminology as applicable under the active rule
Not Assessable	The system cannot make the check reliably because evidence, scale or applicability is insufficient
PDP	Principal display panel
Rule version	Approved machine-readable legal logic with source and effective dates
USP	Unit sale price

Official references

Ref.	Official source	Use in this PRD
R1	Legal Metrology Act, 2009 - India Code	Statutory basis, including pre-packaged commodity declaration and penalty provisions.
R2	Legal Metrology (Packaged Commodities) Rules, 2011 - official consolidated compilation	Core declarations, principal display panel, font/presentation, schedules, exemptions and amendments.
R3	Legal Metrology (Packaged Commodities) Amendment Rules, 2023	Definitions and e-commerce declaration changes used in channel requirements.
R4	Department of Consumer Affairs / PIB - medical device packaging amendment, 2025	Routes location/font dimensions for medical-device packages to Medical Devices Rules, 2017.
R5	Department of Consumer Affairs / PIB - pan masala declaration amendment	Mandatory declaration requirements for all pan masala package sizes from 1 February 2026.
R6	Legal Metrology (Packaged Commodities) First Amendment Rules, 2026	Searchable/sortable country-of-origin filter for imported e-commerce products from 1 July 2026.
R7	Legal Metrology (Packaged Commodities) Third Amendment Rules, 2026	AEO bonded-warehouse declaration and Rule 27 administration changes.
R8	Department of Consumer Affairs / PIB - improvement notice under Jan Vishwas reforms	Improvement-notice workflow and Legal Metrology commencement from 1 May 2026.
R9	Department of Consumer Affairs - 2026 edible-oil standard pack-size SOP amendment	Category-specific package-size and equivalent weight requirements/transition.

Source maintenance

Before each production rule release, the rule administrator must confirm the source against the current official Gazette/Department publication, retain the source file and hash, record the legal interpretation and run the approved regression pack. This source list is a PRD baseline, not a substitute for ongoing legal monitoring.